Release notes: v0.1.0

Repository source: RELEASE_NOTES.md

Release date: September 6, 2026
Author: Bruce J. Swihart
Status: AI-assisted, non-peer-reviewed research draft seeking independent mathematical verification

Canonical repository: github.com/swihart/minvol-degree3-spectral-bound

Versioned release: v0.1.0

Main manuscript: PDF | LaTeX source

Purpose of this release

Version v0.1.0 is the initial public research draft of a proposed degree-3 spectral refinement of Nishioka's lower bound for the three-dimensional Blaschke-Lebesgue problem.

The proposed bound is

$$\mathrm{Vol}(K) \geq \frac{\pi}{1914}\left(259 - 27\sqrt{\frac{15183}{17303}}\right) d^3 \approx 0.3836027047 d^3.$$

The proposed contribution is an independent spectral mechanism: the proof separates the degree-3 curvature energy, derives a new determinant inequality for that component, and uses it to cap the portion of the total curvature budget that can receive Nishioka's degree-3 coefficient.

Included materials

This release contains:

- the integrated LaTeX paper and compiled PDF;
- expanded bridge lemmas and a detailed degree-3 tensor derivation;
- a claim-dependency graph, proof ledger, and internal proof audit;
- exact symbolic Python checks and exhaustive pairing enumeration;
- independent base-R pairing and numerical checks;
- automated GitHub Actions workflows for Python, R, document builds, metadata checks, and PDF preflight;
- reproducibility, clean-clone, licensing, citation, and AI-provenance records; and
- typeset PDF counterparts of all Markdown documentation.

Verification status

The finite algebraic identities and constants are checked with exact symbolic arithmetic. Independent Python and R implementations agree numerically, and the repository has passed clean hosted builds and a recorded clean-clone reproduction.

The v0.1.0 release procedure first requires the exact release-candidate commit to pass the three hosted GitHub Actions jobs on `main` and one final clean-clone test. That same commit is then tagged `v0.1.0`; the tag-triggered workflow must also pass before the GitHub release is published. The release page records the tested commit and the final result.

These checks establish reproducibility of the supplied computations and artifacts. They do not constitute independent subject-matter review or peer review of the complete proof.

Scope and limitations

HYRA publicly claims substantially stronger analytic and computer-assisted bounds by a different method. This release does not claim the strongest currently proposed numerical coefficient. It also does not prove the Meissner conjecture or identify a minimizing body.

Errors, missing hypotheses, normalization issues, and literature references may still be found. Corrections should preserve the v0.1.0 record and be issued under a new version tag.

Preferred citation

Bruce J. Swihart, *A Degree-Three Spectral Refinement of Nishioka's Lower Bound for the Three-Dimensional Blaschke-Lebesgue Problem*, version v0.1.0, public research draft, September 6, 2026.

The machine-readable citation is in `CITATION.cff`.